AI Scenario Simulator

Problem Statement & Use Cases

Digital Learning Transformation Initiative | 2026

Problem Statement

The scenario experiences embedded in today's digital training courses ask learners to read a situation, select one of three or four pre-written response options, and receive templated feedback. The "right answer" is usually apparent before a learner finishes reading the choices. There is no tailoring of the "what would you do" options, limited continuation and consequence ("here's what happens next based on your response, now what?"), templatized feedback and follow-up, and no recognition that real-world situations rarely resolve in a single decision.

This model creates a fundamental gap between training and application. Learning science is clear that behavior change requires more than knowing the right answer — it requires confronting the social pressures, competing motivations, emotional barriers, and moment-to-moment uncertainties that determine whether knowledge becomes action. A student who can correctly identify the "right choice" in a bullying scenario may still hesitate to intervene because they don't believe their action will matter, feel unsupported, or don't know what to say when the situation escalates. A facilities worker who knows lockout/tagout protocol may skip a step under production pressure. A first responder trained in de-escalation may freeze when the scripted scenario they practiced bears little resemblance to the situation in front of them.

Static scenarios cannot pressure-test these behavioral determinants. They produce passive engagement rather than active practice; check-box completion rather than skill development; and generic options and feedback rather than personally relevant inputs and guidance.

The consequences compound across sectors: in commercial settings, learners "click through" scenarios as efficiently as possible; in higher education, students disengage from content they see as boring or irrelevant; in public safety, practitioners complete required training without ever stress-testing the applied judgment they need in the field.

Despite holding thousands of expert-designed courses — each with validated behavioral objectives, sector-specific context, and content purpose-built for high-stakes environments — there is currently no mechanism to translate that depth into scenarios that are **dynamic, conversational, and genuinely adaptive** to how individual learners actually think, hesitate, and (most importantly) act.

Overarching Positioning

Replace static, choice-limited scenario screens with immersive, AI-powered, chat-based experiences that unfold dynamically in response to what learners actually say — pressure-testing the behavioral determinants that training is designed to develop, and delivering feedback that is personalized, instructionally grounded, and adapted to the learner's specific role, sector, requirements, and competency level.

This is not a cosmetic upgrade to existing scenarios. It is a fundamental shift in how applied learning is delivered — from passive selection to **active practice**, from templated feedback to

adaptive dialogue, and from single-moment snapshots to **scenarios that evolve as real situations do**.

Learner Use Case Scenarios

1. Bystander Intervention — When Knowing the Right Answer Isn't Enough (Education)

- A college student taking a sexual misconduct prevention course encounters a scenario: a peer at a party appears intoxicated and is being led away by someone they seem uncomfortable with. The learner types what they would do.
- A K-12 student completing a bullying prevention course is placed in a hallway scene — a classmate is being targeted by a group of older students. What do you do? Why? What stops you?

👉 ***The scenario doesn't accept "say something" as a complete answer. It probes further:***

- Why haven't the others around you acted? Does that change how you see the situation?
- What do you say, specifically? What if the person you're trying to help pushes back?
- What resources or policies at your institution would support you? Do you know how to access them?
- What happens next — and how do you respond if the situation escalates?

Each probe is mapped to a construct from the Individual Determinants of Behavior framework (Intervention Mapping): knowledge, attitudes and beliefs, perceived outcomes, social norms, personal norms, behavioral skills, perceived behavioral control, and cues to action. The scenario continues to unfold based on the learner's responses, potentially resulting in positive or negative outcomes that allow them to revisit earlier decisions and understand their real-world consequences.

👉 ***The same framework applies to harassment disclosure responses, cybersecurity phishing scenarios, and mental health crisis bystander situations — all high-usage topic areas in the education portfolio.***

2. Safety Protocol Under Real-World Pressure (Commercial Sector — AEC & Industrial Manufacturing)

- A facilities worker in an industrial environment notices an unexpected chemical spill near active equipment. They are completing a hazard communication training. The scenario asks: what do you assess first? What steps do you take, and in what order?

- An AEC worker finishing a lockout/tagout course is walked through a situation where a colleague is about to work on a machine that has not been properly de-energized. Production pressure is building. What do you do?
- A worker in a construction setting encounters a situation involving electrical safety or stormwater pollution prevention — the scenario presents competing cues (a rushed supervisor, an unclear protocol, a colleague who skips the step) and asks the learner to talk about how they would navigate it.

👉 ***The scenario unfolds:***

- The learner's response determines what happens next — if they skip a safety step, the scenario surfaces a consequence; if they escalate correctly, the situation resolves
- The AI probes whether the learner knows the correct protocol AND whether they have the behavioral confidence and skills to apply it when facing real-world resistance, time pressure, or social friction
- Feedback is grounded in the course's learning objectives and compliance standards (OSHA, HAZCOM, NFPA, EPA) — not generated from external sources

👉 ***These scenarios address a core Commercial engagement problem: learners who see safety training as a checkbox exercise are asked to demonstrate, in their own words, that they actually know what to do. Passive clicking is no longer sufficient, and the engagement results in highly tailored and relevant feedback that strengthens their ability to do their job safely and/or effectively (an important value driver for their organization).***

3. High-Stakes Field Judgment (Public Sector — First Responders)

- A law enforcement officer completing a de-escalation course encounters a scenario involving a person in apparent mental health crisis. What do you assess? What do you say? How do you adapt when the person's behavior changes?
- A dispatcher or EMS worker responding to a scenario simulating an ambiguous incoming call — the situation may involve an active threat, a medical emergency, or a mental health crisis — must triage in real-time by typing their assessment and intended response.
- A firefighter in a HazMat training scenario is guided through a rapidly evolving incident: initial assessment, communication to command, containment decisions, and response to new information. Each response triggers the next phase of the scenario.

👉 ***Critical safeguards apply:***

- The AI is trained to recognize responses that fall outside the domain of the learning material and respond appropriately — not attempt to guide the learner through situations or answer questions beyond its appropriate scope
- Scenario depth is calibrated to accreditation requirements (OSHA, NFPA, state licensing bodies) and role-specific competency standards
- Feedback is tied to professional standards and defensible course content — not synthesized from the open internet

Tech / Experience Considerations

Scenario Authoring Tool (Internal)

- A dedicated tool for the Knowledge & Accreditations team to generate, review, manage, and approve scenario templates and prompt sequences — drawing from existing course content, learning objectives, and the existing scenario bank as a starting repository
- LEDs should be able to build scenario prompts aligned to specific behavioral objectives, configure escalation thresholds, and verify AI output quality before scenarios go live in courses
- Authoring interface should minimize the learning curve — the tool should guide scenario creation rather than require technical expertise
- The design of the experience should account for scenarios that are presented via text, static images, and videos
- Scenarios should be auto-generated, then allow the SME & LED to validate – If these take too long to create, the LED team will not use them due to production time constraints

AI Model Constraints & Accuracy

- The AI must be trained exclusively on expert-created content, learning objectives, approved metadata, and inputs from training administrators (resources, policies, SOPs, etc.) — it cannot synthesize responses from external internet sources
- “Correct” responses must be defined by LEDs or authoring admins (if customer facing). The tool must be intelligent enough to “score” a learner’s response to scenarios against a set of criteria provided by LEDs and admins without the criteria needing to cover every possible iteration of phrasing in the responses.This has been a challenging factor for K&A in recent gamified Commercial content.
- Responses must be 100% accurate, non-drifting, and grounded in the course content they accompany. If the learner inputs a response that cannot be easily mapped to “correct answer” we need to be able to fall back to a more multiple choice pathing.
- Escalation protocols must be built in: when a learner’s response triggers recognized crisis indicators (mental health emergency language, active threat disclosures, reports of imminent harm), the system escalates appropriately rather than continuing the scenario
- Boundary behavior must be defined upfront: learners should not be able to take a scenario down a path the system cannot responsibly support
- Opt-out capability: organizations and administrators must be able to disable conversational AI scenarios for specific courses, learner populations, or deployments — to address ethical concerns, data sensitivity requirements, or regulatory constraints that restrict AI-generated interactions in certain contexts. May want to allow for learners themselves to opt out of AI-centric experiences. - Need to also provide Accessibility support or an accessible version of the scenario.

Auditability & Compliance

- Full learner-level audit trail: the complete chain of scenario prompts and learner responses must be logged and accessible to administrators for compliance reviews, accreditation audits, or legal subpoena
- Scores and feedback must be traceable back to specific behavioral objectives and scenario constructs
- Data retention and access policies must be clearly defined and defensible
- Regulatory procurement context: institutional buyers are increasingly required to evaluate AI tools against formal policy frameworks before purchase. As a reference point, the SUNY System-wide AI Policy (April 2026) illustrates how higher education regulators are defining requirements for AI in learning environments and what vendors must be prepared to address during procurement review: https://www.suny.edu/about/leadership/board-of-trustees/meetings/webcastdocs/Reso_SUNYSystemwideAIPolicy_April2026.pdf
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Data Privacy & Learner Insights

- Learner response data must be handled with strict privacy protections — particularly for minors (K-12), where data governance and parental consent considerations apply
- Aggregate, anonymized theme data from learner responses should be available to training administrators to identify patterns (e.g., which concepts or situations learners consistently struggle with) for program planning and engagement strategy
- Individual learner data must never be exposed through aggregate reporting (unless required by law, or if the learner consents)

Seat Time & Input Modality

- Conversational write-in responses will take meaningfully more time than selecting from 4 static options — scenario design must account for this, and seat time implications should be managed proactively
- Voice/narrated input should be supported as an alternative to typing, reducing friction and making the experience more natural — particularly for field-based, mobile-first, or time-constrained learners; and a baseline accessibility requirement for certain Commercial sector workforces where learners may have limited literacy, limited English proficiency, or educational backgrounds at or below an 8th grade level — some employers in this segment specifically hire workers who cannot read or write proficiently, or are non-native English speakers, making narrated and spoken-response interfaces essential rather than optional
- Where existing course content is replaced or restructured to make room for the scenario simulator, overall module length should be managed to maintain comparable total seat time

LMS Integration & Completion Tracking

- Scenario completion must integrate with existing LMS completion and credit tracking workflows — learners who complete the scenario simulator progress in the same way they currently do for static scenario screens
- For accredited trainings with prescribed assessment or scenario requirements, the simulator must be configurable to meet those specifications (while still enabling the dynamic experience, if allowable) or completely locked down as needed.

Future Considerations

- Standalone scenario-based training modalities for ongoing practical skill refreshers — deployed independently of full courses for continuous learning and periodic competency maintenance, assignable in lieu of a full course, or opted into directly by learners absent a formal training assignment
- Customer-configurable scenario customization — allowing organizations to inject their own policies, resources, roles, and situational context into scenarios to maximize real-world relevance; extended to include customer-facing capabilities to adapt existing scenarios or create new scenarios from scratch using their own subject matter expertise, policies, and organizational context
- Integration with AI Aptitude Assessment to create a continuous loop: scenario performance data informs competency profiles; competency gaps inform which scenarios learners are directed to next; over time, Aptitude Assessment can augment or replace traditional quizzes, capturing and reporting on true demonstrated capabilities rather than test-taking ability
- Adaptive scenarios based on learner or organizational context — automatically adjusting scenario circumstances based on factors such as job role (desk-based vs. manufacturing line worker), learning environment (commuter vs. residential student), or prior performance history, so that the experience is directly relevant to the learner's real-world situation
- Proactive training reinforcement — practice-based microlearning teed up proactively based on identified learner skill gaps, delivered in the moments between formal training cycles when behavioral application is most likely to occur
- Group-based scenario experiences that enable coordinated social learning, peer discussion, and human connection — allowing teams or cohorts to work through shared scenarios together, compare responses, and reflect on divergent judgment calls
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Value Proposition

Transform in-course scenarios from passive knowledge checks into dynamic, immersive practice environments — allowing for learner-driven inputs and delivering personalized, instructionally grounded feedback that pressure-tests the knowledge, attitudes, and skills training is actually designed to develop, and producing engagement that learners and organizations recognize as genuinely valuable.

Right to Win (Strengthened & Differentiated)

The competitive advantage is not the ability to build a conversational AI scenario engine — it is the depth of validated content, behavioral design expertise, and sector-specific trust that makes the scenarios those learners encounter rigorous, defensible, and genuinely relevant to their lives and roles.

1. Expert-Created, Accredited Content Catalogue

- Thousands of courses purpose-built for education, commercial, and public sector learners — each with validated learning objectives, compliance alignment, and role-specific context
- Every scenario the AI can draw on is grounded in expert-created, legally compliant, accredited content — not synthesized from the open internet
- An existing scenario bank from current courses provides a ready-made starting repository for scenario authoring

2. Behavioral Science Foundation (Intervention Mapping)

- The Individual Determinants of Behavior framework — spanning knowledge, attitudes and beliefs, perceived outcomes, social norms, personal norms, behavioral skills, perceived behavioral control, and cues to action — provides the instructional architecture for scenario design
- Scenarios are not conversation starters; they are structured probes of the specific behavioral determinants that determine whether a skill gets applied in real life
- This rigor differentiates the experience from generic AI role-play and grounds it in decades of behavioral change research

3. Learning Experience Design Expertise

- In-house LEDs with sector-specific expertise who can build scenario prompts, rubrics, and escalation logic that are instructionally sound and contextually appropriate
- The scenario authoring tool is designed to be operated by this team — ensuring that every scenario that reaches a learner has been designed and reviewed by a qualified professional
- Competency in translating compliance and accreditation requirements into scenario experiences that satisfy both pedagogical and regulatory standards

4. Sector-Specific Context at Scale (25+ Years)

- Deep understanding of the regulatory, compliance, learning and behavioral science, and cultural contexts that shape what “realistic” and “defensible” mean across K-12, higher education, AEC, industrial manufacturing, fire, law enforcement, EMS, and dispatch environments
- Existing customer relationships and institutional trust that a generic AI scenario tool cannot replicate
- Content metadata including industry, role, job level, and accreditation tags that enable hyperpersonalized scenario tailoring in near real-time

5. Closed-Loop Ecosystem Integration

- Scenario performance data can connect to AI Aptitude Assessment competency profiles to identify behavioral gaps and direct learners to targeted practice
- The AI Knowledge Assistant can provide performance support in the moments between scenarios and real-world application
- Together, these three initiatives create a continuous loop: learn → simulate → assess → support — with data flowing between each stage

Strategic Positioning (Clear Contrast with Current and Generic Alternatives)

Today's Static Scenario Screen:

- 3–4 pre-written options with an obvious correct answer
- Templated feedback that does not adapt to what the learner actually said or thought
- No continuation, no consequence, no opportunity to practice the behavioral skills training is supposed to build
- Passive engagement that reinforces check-box behavior rather than disrupting it

A Generic AI Role-Play Tool:

- Can generate conversational scenarios but lacks grounding in validated course content or learning objectives
- No behavioral science architecture — conversations may be engaging but are not systematically designed to surface and develop specific determinants of behavior
- No sector-specific compliance alignment; no escalation protocols for high-risk inputs; no audit trail
- Not defensible for accreditation, licensing, or compliance use cases

The AI Scenario Simulator:

Grounded in expert-created, accredited content, architected around behavioral science, hyperpersonalized to role and sector, built with defensible escalation and audit capabilities, and designed to convert passive training completion into active, applied skill practice that learners and administrators recognize as genuinely impactful.

Foundation for Future Transformation

This initiative is not just an upgrade to a course component — it lays the groundwork for:

- Standalone scenario-based training as an ongoing practical skill modality — deployed independently for continuous learning, periodic refreshers, and competency maintenance between formal training cycles, with individual experiences assignable in lieu of a full course or opted into by learners directly absent a formal training assignment

- Customer-configurable scenario experiences — organizations inject their own policies, roles, job titles, and situational context to make scenarios hyper-relevant to their specific environment; extended to include customer-facing capabilities to adapt existing scenarios or build new ones from their own subject matter expertise and organizational context
- Aggregate behavioral insight reporting — anonymized data on how learner populations respond to key scenarios informs organizational training strategy and helps identify where behavioral gaps are concentrated
- Integration with AI Aptitude Assessment — scenario performance feeds into competency profiles; profiles drive personalized scenario assignments; Aptitude Assessment can augment or replace traditional quizzes to capture and report on true demonstrated capabilities rather than test-taking performance alone
- Integration with AI Knowledge Assistant — closing the loop between in-training practice and real-world performance support; Scenario Simulator serves as a test-and-learn foundation for Knowledge Assistant, building organizational fluency with conversational AI in training and earning the credibility needed to move faster on that greater-complexity initiative
- Foundation for fully adaptive, AI-driven learning pathways where scenario complexity, content depth, and feedback sophistication adjust dynamically based on demonstrated competency
- Adaptive scenarios based on learner and organizational context — scenario circumstances adjust dynamically based on job role (desk-based vs. manufacturing line), learning environment (commuter vs. residential student), or demonstrated performance history, ensuring every learner faces a scenario directly relevant to their real-world situation
- Proactive training reinforcement — practice-based microlearning surfaced proactively in response to identified skill gaps, delivered in the moments between formal training when behavioral application is most likely to occur
- Group-based scenario experiences that support coordinated social learning, peer reflection, and human connection — allowing cohorts and teams to work through shared scenarios together and compare their judgment and responses
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👉 ***In other words:***

We move from “scenario as assessment” → “scenario as the most powerful skill-building moment in the course.”